Test Series Code

08

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 15/03/2026

ANSWER KEY

1.	(4)	46.	(2)	91.	(1)	136.	(3)
2.	(1)	47.	(3)	92.	(4)	137.	(3)
3.	(2)	48.	(2)	93.	(3)	138.	(3)
4.	(2)	49.	(2)	94.	(1)	139.	(3)
5.	(2)	50.	(3)	95.	(1)	140.	(1)
6.	(2)	51.	(1)	96.	(3)	141.	(1)
7.	(3)	52.	(1)	97.	(2)	142.	(2)
8.	(1)	53.	(4)	98.	(4)	143.	(2)
9.	(2)	54.	(1)	99.	(2)	144.	(3)
10.	(3)	55.	(4)	100.	(4)	145.	(3)
11.	(2)	56.	(2)	101.	(3)	146.	(2)
12.	(3)	57.	(3)	102.	(1)	147.	(3)
13.	(1)	58.	(2)	103.	(1)	148.	(4)
14.	(2)	59.	(1)	104.	(2)	149.	(2)
15.	(3)	60.	(1)	105.	(3)	150.	(1)
16.	(1)	61.	(2)	106.	(2)	151.	(2)
17.	(1)	62.	(2)	107.	(1)	152.	(2)
18.	(1)	63.	(2)	108.	(1)	153.	(2)
19.	(3)	64.	(2)	109.	(4)	154.	(2)
20.	(1)	65.	(1)	110.	(1)	155.	(4)
21.	(1)	66.	(1)	111.	(2)	156.	(1)
22.	(1)	67.	(1)	112.	(3)	157.	(2)
23.	(2)	68.	(4)	113.	(4)	158.	(4)
24.	(1)	69.	(3)	114.	(1)	159.	(2)
25.	(2)	70.	(3)	115.	(1)	160.	(3)
26.	(2)	71.	(3)	116.	(3)	161.	(2)
27.	(2)	72.	(3)	117.	(2)	162.	(1)
28.	(3)	73.	(1)	118.	(4)	163.	(2)
29.	(2)	74.	(1)	119.	(1)	164.	(2)
30.	(2)	75.	(2)	120.	(3)	165.	(2)
31.	(1)	76.	(3)	121.	(4)	166.	(3)
32.	(2)	77.	(1)	122.	(2)	167.	(3)
33.	(1)	78.	(1)	123.	(3)	168.	(2)
34.	(1)	79.	(2)	124.	(1)	169.	(3)
35.	(1)	80.	(1)	125.	(3)	170.	(4)
36.	(2)	81.	(3)	126.	(2)	171.	(2)
37.	(1)	82.	(2)	127.	(4)	172.	(1)
38.	(3)	83.	(1)	128.	(1)	173.	(3)
39.	(2)	84.	(1)	129.	(3)	174.	(2)
40.	(2)	85.	(1)	130.	(4)	175.	(1)
41.	(1)	86.	(3)	131.	(2)	176.	(2)
42.	(1)	87.	(3)	132.	(4)	177.	(1)
43.	(1)	88.	(2)	133.	(1)	178.	(3)
44.	(2)	89.	(2)	134.	(2)	179.	(1)
45.	(3)	90.	(4)	135.	(3)	180.	(3)

[Contd...

Hints & Solutions

1. (4)

Velocity at midpoint M:

$$v_M = \sqrt{\frac{u^2+v^2}{2}}$$

$$v_M = \sqrt{\frac{2^2+14^2}{2}} = \sqrt{\frac{4+196}{2}}$$

$$= \sqrt{\frac{200}{2}} = \sqrt{100} = 10 \text{ m/s}$$

(Statement A is Correct)

Average Velocity:

$$v_{avg} = \frac{u+v}{2}$$

$$v_{avg} = \frac{2+14}{2} = \frac{16}{2} = 8 \text{ m/s}.$$

(Statement B is Correct)

Ratio of times ($t_{MB} : t_{AM}$):

Time $t = \frac{\Delta v}{a}$.

Since acceleration a is constant, time is proportional to change in velocity.

$$t_{AM} \propto (v_M - v_A) = 10 - 2 = 8$$

$$t_{MB} \propto (v_B - v_M) = 14 - 10 = 4$$

Ratio $\frac{t_{MB}}{t_{AM}} = \frac{4}{8} = \frac{1}{2}$.

(Statement C is Correct)

Total Time (T) if distance $S = 64$ m:

$$S = v_{avg} \times T$$

$$64 = 8 \times T \Rightarrow T = 8 \text{ s}.$$

(Statement D says 10 s, so Statement D is Incorrect).

(NCERT Ed. 2025-26, 11th, Page No. 18)

2. (1)

Between the 1st and 4th drop, there are 3 equal time intervals.

The 1st drop has fallen for 3 intervals (Total height H).

The 2nd drop has fallen for 2 intervals.

Since distance $d \propto t^2$:

$$\frac{d_{2nd}}{H} = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

The 2nd drop has fallen $\frac{4H}{9}$

Height from ground $= H - \frac{4H}{9} = \frac{5H}{9}$

(NCERT Ed. 2025-26, 11th, Page No. 20)

3. (2)

Moment of Inertia (I):

For a thin ring, $I = MR^2$.

$$I = 5 \times (2)^2 = 20 \text{ kg} \cdot \text{m}^2$$

Net Torque (τ): Since the force acts tangentially, the angle between the force and the radius vector is $90°$ ($\sin 90° = 1$).

Torque per nozzle $= F \times R$.

Since there are 4 nozzles:

$$\tau_{net} = 4 \times (10 \text{ N} \times 2 \text{ m}) = 80 \text{ N} \cdot \text{m}$$

Angular Acceleration (α):

Using $\tau = I\alpha$:

$$\alpha = \frac{\tau}{I} = \frac{80}{20} = 4 \text{ rad/s}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 120)

4. (2)

$$T = Mg = 4 \times 10 = 40 \text{ N}$$

$$T = \frac{mv^2}{R}$$

$$40 = \frac{3 \times v^2}{1.5}$$

$$40 \times 1.5 = 3v^2$$

$$60 = 3v^2$$

$$v^2 = 20$$

$$v = \sqrt{20} = 2\sqrt{5} \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 63)

5. (2)

The force of static friction acts on the crate in the forward direction, which is the same as the direction of its displacement. Since the force and displacement are in the same direction (angle is 0°), the work done is positive.

(NCERT Ed. 2025-26, 11th, Page No. 74)

6. (2)

$$\Delta K.E. = \int P(t)dt$$

$$K.E._f = \int_0^6 (2t)dt$$

$$K.E._f = 2\left[\frac{t^2}{2}\right]_0^6$$

[Contd...

$K.E._f = [t^2]_0^6$

$K.E._f = 6^2 - 0^2 = 36$ J

(NCERT Ed. 2025-26, 11th, Page No. 83)

7. (3)

$I_1 = I_{CM} = \frac{ML^2}{12}$

$I_2 = I_{CM} + Md^2$

$I_2 = \frac{ML^2}{12} + M\left(\frac{L}{6}\right)^2$

$I_2 = \frac{ML^2}{12} + \frac{ML^2}{36}$

$I_2 = \left(\frac{3}{36} + \frac{1}{36}\right)ML^2 = \frac{4ML^2}{36} = \frac{ML^2}{9}$

$\text{Ratio} = \frac{\frac{ML^2}{9}}{\frac{ML^2}{12}} = \frac{12}{9} = \frac{4}{3}$

(NCERT Ed. 2025-26, 11th, Page No. 116)

8. (1)

Time of flight for one ball: $T = \frac{2u}{g}$.

Height $H = \frac{u^2}{2g} \Rightarrow 20 = \frac{u^2}{20} \Rightarrow u = 20$ m/s.

Total time of flight $T = \frac{2 \times 20}{10} = 4$ s.

There are 4 balls. To keep them in air, the interval

$\Delta t = \frac{T}{4} = \frac{4}{4} = 1$ s.

(NCERT Ed. 2025-26, 11th, Page No. 18)

9. (2)

Velocity of Ship, $\vec{v}_s = -10\hat{i}$.

Velocity of Wind, $\vec{v}_w = 10\hat{j}$.

The flag flutters in the direction of the wind relative to the ship.

$\vec{v}_{w/s} = \vec{v}_w - \vec{v}_s = 10\hat{j} - (-10\hat{i}) = 10\hat{i} + 10\hat{j}$.

This vector points towards North-East.

(NCERT Ed. 2025-26, 11th, Page No. 31)

10. (3)

Resultant wave

$y = y_1 + y_2 = 4[\sin(2x - 3t) + \sin(2x + 3t)]$

Using $\sin(A+B) + \sin(A-B) = 2\sin A\cos B$:

$y = 2(4)\sin(2x)\cos(3t) = 8\sin(2x)\cos(3t)$.

Amplitude at position x: $A(x) = |8\sin(2x)|$.

At $x = \pi/4$:

$A = 8\sin(2 \times \pi/4) = 8\sin(\pi/2) = 8(1) = 8$ cm.

(NCERT Ed. 2025-26, 11th, Page No. 290)

11. (2)

$T^2 \propto R^3$.

Given $\frac{T_P}{T_Q} = \frac{1}{8}$.

$\frac{R_P}{R_Q} = \left(\frac{T_P}{T_Q}\right)^{2/3} = \left(\frac{1}{8}\right)^{2/3} = \left(\frac{1}{2^3}\right)^{2/3} = \frac{1}{4}$.

Areal velocity is given by $\frac{dA}{dt} = \frac{L}{2m}$.

For a circular orbit, Angular Momentum

$L = mvr = m\left(\sqrt{\frac{GM}{R}}\right)R = m\sqrt{GMR}$.

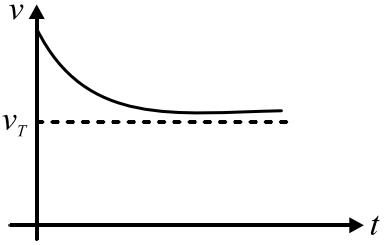
Therefore, Areal Velocity $\frac{dA}{dt} \propto \sqrt{R}$.

Ratio $\frac{v_{A,P}}{v_{A,Q}} = \sqrt{\frac{R_P}{R_Q}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$.

(NCERT Ed. 2025-26, 11th, Page No. 129)

12. (3)

When the body is dropped in a viscous fluid with a velocity greater than terminal velocity then the viscous force acts on the body opposing motion and the speed decreases and finally achieve terminal velocity.



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[Contd...

13. (1)

Heat current H is same in series.

$$\frac{K_A A(T_1 - T)}{L} = \frac{K_B A(T - T_2)}{L}$$

$$2K(100 - T) = K(T - 10)$$

$$200 - 2T = T - 10$$

$$210 = 3T \Rightarrow T = 70°\text{C}$$

(NCERT Ed. 2025-26, 11th, Page No. 216)

14. (2)

According to Wien's Displacement Law:

$\lambda_m T = \text{constant}$.

$\lambda_1 T_1 = \lambda_2 T_2 \Rightarrow \lambda_m(2000) = \lambda_2(3000)$

$\lambda_2 = \frac{2}{3}\lambda_m$.

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15. (3)

Vertical equilibrium: $2T\sin(37°) = mg$

$2T(0.6) = 0.5 \times 10 = 5$

$1.2T = 5 \Rightarrow T = \frac{5}{1.2} = \frac{50}{12} \approx 4.16$ N

Closest option is 4.2 N

(NCERT Ed. 2025-26, 11th, Page No. 58)

16. (1)

$f \propto \frac{1}{L}$ for a closed pipe ($f = \frac{v}{4L}$).

Since $L_B > L_A$, $f_B < f_A$. $f_A - f_B = 4$.

$$\frac{v}{4L_A} - \frac{v}{4L_B} = 4.$$

$$\frac{f_A}{f_B} = \frac{L_B}{L_A} = \frac{30.5}{30} = \frac{61}{60}.$$

Let $f_A = 61k, f_B = 60k$.

$61k - 60k = 4 \Rightarrow k = 4$.

$f_A = 244$ Hz, $f_B = 240$ Hz.

(NCERT Ed. 2025-26, 11th, Page No. 294)

17. (1)

Boyle's Law ($P_1V_1 = P_2V_2$).

At surface: $P_2 = P_{atm} = H\rho g, V_2 = 8V$.

At bottom: $P_1 = P_{atm} + h\rho g = (H + h)\rho g, V_1 = V$.

$(H + h)\rho g \times V = H\rho g \times 8V$.

$H + h = 8H \Rightarrow h = 7H$.

(NCERT Ed. 2025-26, 11th, Page No. 184)

18. (1)

Dimensional formula of Energy $[ML^2T^{-2}]$.

$n_1[M_1L_1^2T_1^{-2}] = n_2[M_2L_2^2T_2^{-2}]$.

$1\text{ J} = 1[\text{kg}][\text{m}]^2[\text{s}]^{-2}$.

$$n_2 = 1 \times \left(\frac{1\text{kg}}{10\text{kg}}\right) \times \left(\frac{1\text{m}}{1000\text{m}}\right)^2 \times \left(\frac{1\text{s}}{60\text{s}}\right)^{-2}.$$

$n_2 = 1 \times (0.1) \times (10^{-6}) \times (3600)$.

$n_2 = 3600 \times 10^{-7} = 3.6 \times 10^{-4}$ units

(NCERT Ed. 2025-26, 11th, Page No. 8)

19. (3)

$u_x = 6, u_y = 8$

$\tan\theta = \frac{u_y}{u_x} = \frac{8}{6} = \frac{4}{3}$.

So $a = \tan\theta = \frac{4}{3}$.

Trajectory: $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$.

$b = \frac{g}{2u_x^2}$ (since $u\cos\theta = u_x$).

$b = \frac{10}{2(6)^2} = \frac{10}{72} = \frac{5}{36}$

(NCERT Ed. 2025-26, 11th, Page No. 39)

20. (1)

$a_c = \frac{v^2}{r} = k^2rt^2 \Rightarrow v^2 = k^2r^2t^2 \Rightarrow v = krt$.

Tangential acceleration $a_t = \frac{dv}{dt} = kr$.

Tangential force $F_t = ma_t = mkr$.

Power $P = F_t \cdot v = (mkr)(krt) = mk^2r^2t$.

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21. (1)

Let mass of full disc be M.

Area $A = \pi R^2$.

Removed mass m. Area $a = \pi\left(\frac{R}{2}\right)^2 = \frac{A}{4}$.

So $m = \frac{M}{4}$.

Let center of original disc be $(0,0)$.

Removed COM is at $(R/2, 0)$.

[Contd...

$$X_{COM} = \frac{M(0) - m(R/2)}{M - m} = \frac{-(M/4)(R/2)}{3M/4}$$

$$= \frac{-R/8}{3/4} = -\frac{R}{6}$$

Shift is $\frac{R}{6}$

(NCERT Ed. 2025-26, 11th, Page No. 98)

22. (1)

Conservation of Energy:

$$-\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{R+h}$$

$$v^2 = gR = \frac{GMR}{R}.$$

$$-\frac{GMm}{R} + \frac{1}{2}m\left(\frac{GM}{R}\right) = -\frac{GMm}{R+h}$$

$$-\frac{1}{R} + \frac{1}{2R} = -\frac{1}{R+h}$$

$$-\frac{1}{2R} = -\frac{1}{R+h} \Rightarrow 2R = R + h \Rightarrow h = R$$

(NCERT Ed. 2025-26, 11th, Page No. 135)

23. (2)

Young's Modulus $Y = \frac{FL}{Al}$

$$l = \frac{FL}{AY}$$

For second wire: $l' = \frac{(2F)(2L)}{(2A)Y} = \frac{4FL}{2AY} = 2\frac{FL}{AY} = 2l$

(NCERT Ed. 2025-26, 11th, Page No. 170)

24. (1)

$V_{max} = \omega A = 10$ cm/s

Given $A = 4$ cm, so $\omega = 10/4 = 2.5$

Velocity at x: $v = \omega\sqrt{A^2 - x^2}$

$5 = 2.5\sqrt{16 - x^2} \Rightarrow 2 = \sqrt{16 - x^2}$

$4 = 16 - x^2 \Rightarrow x^2 = 12 \Rightarrow x = \sqrt{12} = 2\sqrt{3}$ cm

(NCERT Ed. 2025-26, 11th, Page No. 268)

25. (2)

$\Delta P = 140 - (-10) = 150$

$\Delta C = 100$

Ratio: $\frac{\Delta P}{\Delta C} = \frac{150}{100} = 1.5$ (This means $1°C = 1.5°P$).

$$S_P = \frac{S_{SI}}{1.5}$$

$$S_P = \frac{900}{1.5} = 600$$

(NCERT Ed. 2025-26, 11th, Page No. 203)

26. (2)

$\vec{v}_r = -30\hat{k}, \vec{v}_w = -10\hat{j}$

$\vec{v}_{r/w} = \vec{v}_r - \vec{v}_w = +10\hat{j} - 30\hat{k}$

Angle with vertical ($\hat{k}$):

$$\tan\theta = \frac{v_{horizontal}}{v_{vertical}} = \frac{10}{30} = \frac{1}{3}$$

$\theta = \tan^{-1}(1/3)$ with vertical

(NCERT Ed. 2025-26, 11th, Page No. 31)

27. (2)

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

Initial: $v = \sqrt{\frac{3RT}{M_{H_2}}}$

Final: Temp $2T$, Molar mass $M_H = M_{H_2}/2$.

$$v' = \sqrt{\frac{3R(2T)}{M_{H_2}/2}} = \sqrt{4\frac{3RT}{M_{H_2}}} = 2v.$$

(NCERT Ed. 2025-26, 11th, Page No. 250)

28. (3)

When pushing at an angle, the vertical component of force adds to weight ($N = mg + F\sin\theta$), increasing limiting friction μN.

When pulling, vertical component reduces normal reaction ($N = mg - F\sin\theta$), reducing friction.

(NCERT Ed. 2025-26, 11th, Page No. 61)

29. (2)

$v = \sqrt{T/\mu}$

$\mu = 0.005$ kg / 1 m $= 0.005$

$v = \sqrt{8/0.005} = \sqrt{1600} = 40$ m/s

Fundamental freq $f = v/2L = 40/(2\times1) = 20$ Hz.

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[Contd...

30. (2)

$m\vec{u} = m\vec{v}_1 + m\vec{v}_2$

$\vec{u} = \vec{v}_1 + \vec{v}_2$

$u^2 = v_1^2 + v_2^2 + 2\vec{v}_1 \cdot \vec{v}_2$

$\frac{1}{2}mu^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2$

$u^2 = v_1^2 + v_2^2$

$(v_1^2 + v_2^2) = v_1^2 + v_2^2 + 2v_1v_2\cos\theta$

$0 = 2v_1v_2\cos\theta$

$\cos\theta = 0 \Rightarrow \theta = 90°$

(NCERT Ed. 2025-26, 11th, Page No. 85)

31. (1)

L.C. = 0.01 mm

Total reading = MSR + CSR

= (3 mm) + 25 × (L.C.)

= (3 mm) + 25 × (0.01 mm) = 3.25 mm

(NCERT Ed. 2025-26, 11th, Page No. 11)

32. (2)

$T_1 = 727 + 273 = 1000$ K

$T_2 = 27 + 273 = 300$ K

$\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{1000} = 1 - 0.3 = 0.7 = 70\%$.

(NCERT Ed. 2025-26, 11th, Page No. 238)

33. (1)

T_{max} (bottom) $= \frac{mv_b^2}{L} + mg$

T_{min} (top) $= \frac{mv_t^2}{L} - mg$

Energy: $v_b^2 = v_t^2 + 4gL$

$T_{max} = \frac{m}{L}(v_t^2 + 4gL) + mg = T_{min} + 2mg + 4mg = T_{min} + 6mg$

Given $T_{max} = 4T_{min}$

$4T_{min} = T_{min} + 6mg \Rightarrow 3T_{min} = 6mg \Rightarrow T_{min} = 2mg$

$\frac{mv_t^2}{L} - mg = 2mg \Rightarrow \frac{v_t^2}{L} = 3g \Rightarrow v_t = \sqrt{3gL}.$

$v_t = \sqrt{3 \times 10 \times 10/3} = \sqrt{100} = 10$ m/s

(NCERT Ed. 2025-26, 11th, Page No. 79)

34. (1)

Rise in temperature

$\Delta t = 20°$C

Time period $T \propto \sqrt{L}$.

$\frac{\Delta\tau}{\tau} = \frac{1}{2}\alpha\Delta\theta$

$\Delta\tau = \left(\frac{1}{2} \times 10^{-5} \times 20\right) \times (24 \times 3600)$

$\Delta\tau = 10^{-4} \times 86400 = 8.64$ s

As temp increases, length increases, time period increases (clock runs slow/loses time).

(NCERT Ed. 2025-26, 11th, Page No. 205)

35. (1)

Bulk Modulus $B = -\frac{\Delta P}{\Delta V / V}$

$\left|\frac{\Delta V}{V}\right| = 1\% = 0.01$.

$\Delta P = 2 \times 10^7$.

$B = \frac{2 \times 10^7}{0.01} = 2 \times 10^9$ Pa.

Compressibility

$k = \frac{1}{B} = \frac{1}{2 \times 10^9} = 0.5 \times 10^{-9} = 5 \times 10^{-10}$ m²/N

(NCERT Ed. 2025-26, 11th, Page No. 173)

36. (2)

New Voltage: $V' = V + 0.5V = 1.5V$

New Current: $I' = I - 0.25I = 0.75I$

New Resistance: $R' = \frac{1.5V}{0.75I} = 2\left(\frac{V}{I}\right) = 2R$

Percentage Change: $\frac{2R - R}{R} \times 100\% = 100\%$

(NCERT Ed. 2025-26, 11th, Page No. 6)

37. (1)

$g' = g \times \left(\frac{r}{R}\right)$

Given that the distance from the center is $r = \frac{R}{6}$:

$g' = g \times \left(\frac{R/6}{R}\right)$

$g' = \frac{g}{6}$

(NCERT Ed. 2025-26, 11th, Page No. 134)

[Contd...

38. (3)

$a = -\omega^2 x$

Slope $= \omega^2 = \dfrac{20}{0.20} = 100$

$\omega = 10$ rad/s

(NCERT Ed. 2025-26, 11th, Page No. 266)

39. (2)

Work done by conservative force depend on initial and final position only. K.E. depend on reference frame.

(NCERT Ed. 2025-26, 11th, Page No. 78)

40. (2)

$P_{gas} = P_{atm} + h\rho g$

Expressing in terms of height of Hg column:

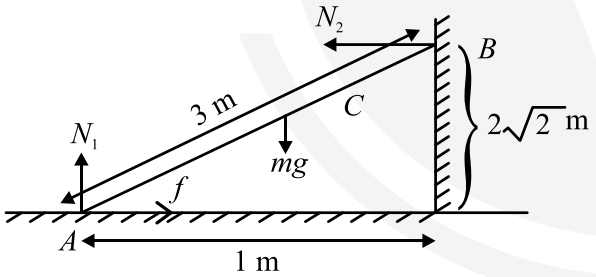
P_{gas} (in cm Hg) $= P_{atm}$ (in cm Hg) $+ h$

Since the open end (right) is higher, the gas pressure is pushing harder.

$P_{gas} = 76 + 10 = 86$ cm of Hg.

(NCERT Ed. 2025-26, 11th, Page No. 184)

41. (1)



From the FBD of ladder:

For translational equilibrium

$N_1 = mg = 200$ N

And $f = N_2$

For rotational equilibrium (About A)

$mg\left(\dfrac{1}{2}\right) = N_2(2\sqrt{2})$

$\dfrac{200}{4\sqrt{2}} = N_2$

$\therefore$ $N_2 = 35.25$ N

and $N_1 = 200$ N

(NCERT Ed. 2025-26, 11th, Page No. 113)

42. (1)

$\Delta U = C_v \Delta T = \dfrac{3}{2} R\Delta T$

Work Done (W):

Given $P = kV^2$.

Using $PV = RT \Rightarrow (kV^2)V = RT \Rightarrow kV^3 = RT$.

Differentiating both sides: $3kV^2 dV = RdT$.

Since $P = kV^2$, we can write:

$3PdV = RdT \Rightarrow PdV = \dfrac{R}{3} dT$.

$W = \int PdV = \int \dfrac{R}{3} dT = \dfrac{1}{3} R\Delta T$

$Q = \Delta U + W$

$Q = \dfrac{3}{2} R\Delta T + \dfrac{1}{3} R\Delta T = \left(\dfrac{9+2}{6}\right) R\Delta T = \dfrac{11}{6} R\Delta T$

(NCERT Ed. 2025-26, 11th, Page No. 230)

43. (1)

Upward Acceleration (A): The pseudo force acts downwards. Net force equation:

$N - mg = ma \Rightarrow N = m(g + a)$ (A $\rightarrow$ II)

Downward Acceleration (B): The pseudo force acts upwards. Net force equation:

$mg - N = ma \Rightarrow N = m(g - a)$. (B $\rightarrow$ IV)

Constant Velocity (C): Acceleration $a = 0$.

Net force equation: $N - mg = 0 \Rightarrow N = mg$.

(C $\rightarrow$ I)

Free Fall (D): The lift falls with $a = g$ downwards.

$N = m(g - g) = 0$ (Weightlessness). (D $\rightarrow$ III)

(NCERT Ed. 2025-26, 11th, Page No. 65)

44. (2)

The property of matter by virtue of which a body tends to regain its original shape and size after removal of deforming force is called as elasticity.

(NCERT Ed. 2025-26, 11th, Page No. 167)

45. (3)

$$\frac{(v_e)_p}{(v_e)_e} = \frac{\sqrt{\dfrac{2GM_p}{R_p}}}{\sqrt{\dfrac{2GM_e}{R_e}}} = \sqrt{\frac{M_p}{M_e} \times \frac{R_e}{R_p}}$$

$$= \sqrt{\frac{8M_e}{M_e} \times \frac{R_e}{R_e/8}} = 8$$

$\therefore$ $(v_e)_p = 8 \times (v_e)_e = 8 \times 11 = 88$ km/s

(NCERT Ed. 2025-26, 11th, Page No. 136)

[Contd...

46. (2)

Seventh period starts with 7s, then fills 5f and 6d, ending with 7p as per Aufbau order.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 62)

47. (3)

First ionization enthalpy decreases down a group; alkali metals are lowest, so Cs (Z=55) minimum among the given atomic numbers.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 87)

48. (2)

Bond order rises when antibonding electrons decrease; thus NO^+ highest, then NO, then NO^-, while peroxide is lowest of all.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 129)

49. (2)

NO_2^- and SO_2 are both bent; NH_4^+ and BF_4^- are tetrahedral, whereas others differ in geometry so only A, B.

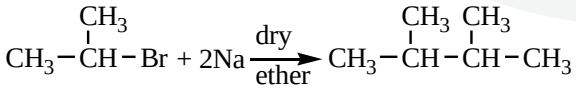
(NCERT Ed. 2025-26, 11th, Part-I Page No. 119)

50. (3)

In free expansion into vacuum, external pressure is zero so w=0; insulated gives q=0, hence $\Delta U=0$ and $\Delta T=0$ for ideal.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 142)

51. (1)



(NCERT Ed. 2025-26, 11th, Part-II Page No. 301)

52. (1)

Lewis bases donate a lone pair; BF_3 has no lone pair on boron, so it cannot donate to any acceptor.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 192)

53. (4)

Atomic radius increases and ionization enthalpy decreases from Be to Ba, so A,B correct; C,D reversed trends for these properties.

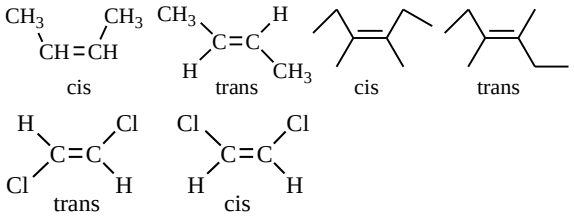
(NCERT Ed. 2025-26, 11th, Part-I Page No. 87)

54. (1)

Bond energy increases with shorter, stronger bond; halogen size grows down group, so H–F strongest then H–Cl, H–Br, H–I finally.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 108)

55. (4)



(NCERT Ed. 2025-26, 11th, Part-II Page No. 308)

56. (2)

Electron affinity is higher for smaller atoms except anomalies; Cl > Br, S > O, and Na > K overall.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 89)

57. (3)

Lyman	UV region
Balmer	Visible region
Paschen	Infrared region

(NCERT Ed. 2025-26, 11th, Part-I Page No. 46)

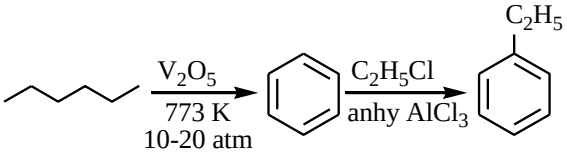
58. (2)

S character increases

Acidity increases

(NCERT Ed. 2025-26, 11th, Part-II Page No. 315)

59. (1)



(NCERT Ed. 2025-26, 11th, Part-II Page No. 322)

60. (1)

- In Bohr's model, the electron's angular momentum in a stationary orbit satisfies $mvr = n\hbar$.
- A fixed circular orbit assigns definite r and p simultaneously, conflicting with Heisenberg uncertainty.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 45)

[Contd...

61. (2)

Carbon forms 40% of glucose; 24g carbon implies 60 glucose, giving 1/3 mol in 0.14 kg water; molality 50/21.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 23)

62. (2)

Nonmetals are more electronegative and form ionic compounds with very reactive metals; they also have higher ionisation enthalpy generally hence.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 187)

63. (2)

Na^+ , Mg^{2+} and Al^{3+} are isoelectronic; increasing nuclear charge pulls electrons closer, so radius decreases: $Mg^{2+} > Al^{3+}$ values are 0.72, 0.54.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 86)

64. (2)

Steric number decides hybridisation: carbonate and BF_3 have three regions so sp^2, while BrO_2^- and BrO_3^- have four so sp^3.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 119)

65. (1)

Since $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$, comparison gives $A = \Delta H^\circ$ and $B = \Delta S^\circ$; endothermic requires positive enthalpy regardless of B, because it represents entropy change.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 183)

66. (1)

Huge K_c favors products, but catalyst affects rate not equilibrium; pressure favors fewer moles, and exothermic K decreases with .

(NCERT Ed. 2025-26, 11th, Part-I Page No. 178)

67. (1)

ClO^- , H_2O_2 , I_2 and Cu^+ each contain intermediate oxidation states, allowing simultaneous oxidation and reduction into two products readily therefore.

(NCERT Ed. 2025-26, 11th, Part-II Page No. 244)

68. (4)

Basic character increases down group as metallic nature rises; Tl_2O_3 is most ionic, hence strongest basic oxide among given options.

69. (3)

Hyperconjugation is a permanent σ-delocalisation effect; increasing alkyl groups gives more βC-H overlaps, stabilising carbocation strongly therefore II true.

(NCERT Ed. 2025-26, 11th, Part-II Page No. 277)

70. (3)

Adsorption chromatography uses solid polar adsorbents; silica gel and alumina provide surfaces, while quicklime or magnesia aren't standard here typically.

(NCERT Ed. 2025-26, 11th, Part-II Page No. 282)

71. (3)

Kjeldahl converts organic nitrogen to ammonium only for amines/amides; nitro nitrogen resists reduction, so fails in this method completely here.

(NCERT Ed. 2025-26, 11th, Part-II Page No. 287)

72. (3)

Strong $-M/-I$ groups deactivate ring and withdraw electron density from ortho/para positions, making meta substitution favored for all these groups.

(NCERT Ed. 2025-26, 11th, Part-II Page No. 276)

73. (1)

Only $2,2,4$ -trimethylpentane contains exactly one secondary CH_2 group; all other carbons are primary or tertiary there giving two secondary hydrogens.

(NCERT Ed. 2025-26, 11th, Part-II Page No. 257)

74. (1)

Carbon lacks d orbitals, limiting covalence; heavier elements rarely form $p\pi - p\pi$ bonds, and carbon shows negative states in carbides too.

75. (2)

Freezing and adsorption increase order, lowering entropy; ammonia formation reduces gas moles, so entropy drops; dissolution increases it instead always.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 158)

[Contd...

76. (3)

I in IO_3^- reduces from +5 to 0 needing five electrons; five iodides oxidize, satisfying iodine balance exactly so $x = 5$.

(NCERT Ed. 2025-26, 11th, Part-II Page No. 246)

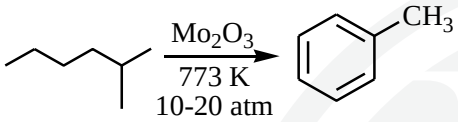
77. (1)

For $A_2B_3 \rightleftharpoons 2A^+ + 3B^-$,

$K_{eq} \approx (2c\alpha)^2(3c\alpha)^3/(c) = 108c^4\alpha^5$, hence α expression equals the fifth-root of K_{eq} over $108c^4$ only here.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 204)

78. (1)



(NCERT Ed. 2025-26, 11th, Part-II Page No. 304)

79. (2)

H-bonding requires H attached to N/O/F; o-nitrophenol is intramolecular, HF intermolecular; state affects strength; properties change in solids and liquids.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 131)

80. (1)

$He < Xe < Ar \approx Kr < Ne$

(NCERT Ed. 2025-26, 11th, Part-I Page No. 90)

81. (3)

Radial nodes make probability density zero at nonzero r; 2s has one node, unlike $1s, 2p, 3d$ in hydrogen atom at finite.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 57)

82. (2)

Reaction: $3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2$;

3 mol phosphate gives 1.5 mol product.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 20)

83. (1)

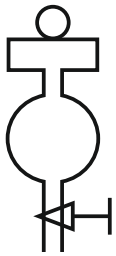
Average mass 35.2 gives $35 + 2x = 35.2$, so $x = 0.1$ of ^{37}Cl; ratio $35:37 = 9:1$ since 0.9 is for ^{35}Cl and 0.1 for ^{37}Cl.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 17)

84. (1)

Down heavy p-block, ns^2 electrons stay inert due to poor shielding and relativistic contraction, stabilising lower oxidation state like $Pb(II)$.

85. (1)



is separating funnel

(NCERT Ed. 2025-26, 11th, Part-II Page No. 281)

86. (3)

In $N_2 + O_2 \rightarrow 2NO$, nitrogen oxidises 0 to +2 and oxygen reduces 0 to –2, hence redox other options are precipitation neutralisation.

(NCERT Ed. 2025-26, 11th, Part-II Page No. 236)

87. (3)

0.2 M 100 mL CH_3COOH = 0.02 mol

50 mL × 0.2 M NaOH = 0.01 mol

Hence buffer.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 202)

88. (2)

For $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, $\Delta_n = +1$

so $K_p = K_c(RT)^1$; thus $\frac{K_p}{K_c} = RT$ at temperature T because moles gas increase from one to two on dissociation.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 178)

89. (2)

Let CH_4 = x and C_2H_4 = y: x + y = 12, x + 2y = 24 giving y = 12, L = 0.5 mol; heat = 0.5 × 1400 = 700 kJ using 24 L molar volume at 25°C approximately.

(NCERT Ed. 2025-26, 11th, Part-I Page No. 156)

90. (4)

In A and B due to conjugation resonance is possible.

(NCERT Ed. 2025-26, 11th, Part-II Page No. 276)

91. (1)

In studying and understanding the physiological and behavioural processes, one can take a physico-chemical approach and use cell-free systems to investigate. This approach enables us to describe the various processes in molecular terms. The approach is established by analysis of living tissues for elements and compounds. It will tell us what types of organic compounds are present in living organisms. In the next stage, one can ask the question: What are these compounds doing inside a cell? And, in what way they carry out gross physiological processes like digestion, excretion, memory, defense, recognition, etc. In other words we answer the question, what is the molecular basis of all physiological processes? It can also explain the abnormal processes that occur during any diseased condition. This physico-chemical approach to study and understand living organisms is called 'Reductionist Biology'. The concepts and techniques of physics and chemistry are applied to understand biology.

(NCERT Ed. 2025-26, 11th, Page No. 85)

92. (4)

During zygotene, chromosomes start pairing together and this process of association is called synapsis. Crossing over is the exchange of genetic material between two homologous chromosomes that takes place in pachytene. The beginning of diplotene is recognised by the dissolution of the synaptonemal complex and the tendency of the recombined homologous chromosomes of the bivalents to separate from each other except at the sites of crossovers. These X-shaped structures, are called chiasmata. The final stage of meiotic prophase I is diakinesis. This is marked by terminalisation of chiasmata.

(NCERT Ed. 2025-26, 11th, Page No. 126)

93. (3)

In monocotyledonous plants, the primary root is short lived and is replaced by a large number of roots. These roots originate from the base of the stem and constitute the fibrous root system.

(NCERT Ed. 2025-26, 11th, Page No. 58)

94. (1)

Hatch and Slack (C_4) pathway: The primary CO_2 acceptor is a 3-carbon molecule phosphoenol pyruvate (PEP) and is present in the mesophyll cells. It performs carboxylation of atmospheric CO_2.

(NCERT Ed. 2025-26, 11th, Page No. 146)

95. (1)

Plant families like Convolvulaceae, Solanaceae are included in the order Polymoniales mainly based on the floral characters.

(NCERT Ed. 2025-26, 11th, Page No. 7)

96. (3)

If fatty acids were to be respired they would first be degraded to acetyl CoA and enter the pathway.

(NCERT Ed. 2025-26, 11th, Page No. 162)

97. (2)

Dmitri Ivanowsky (1892) recognised certain microbes as causal organism of the mosaic disease of tobacco. M. W. Beijerinck named the new pathogen "virus". Viruses are obligate parasites. W.M. Stanley (1935) showed that viruses could be crystallised and crystals consist largely of proteins. In general, viruses that infect plants have single stranded RNA. Bacterial viruses or bacteriophages (viruses that infect the bacteria) are usually double stranded DNA viruses.

(NCERT Ed. 2025-26, 11th, Page No. 20)

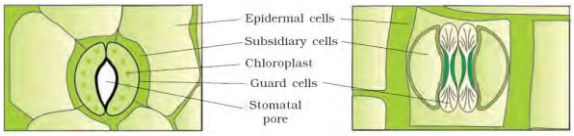
98. (4)

The form and size of algae is highly variable, ranging from colonial forms like *Volvox* and the filamentous forms like *Ulothrix* and *Spirogyra*. A few of the marine forms such as kelps, form massive plant bodies. *Ectocarpus* is a simple branched, filamentous brown alga. The food is stored as floridean starch in red algae (*Porphyra* is red alga).

(NCERT Ed. 2025-26, 11th, Page No. 24)

99. (2)

The given diagram represents the stomatal apparatus in a dorsiventral leaf. Each stoma is composed of two bean-shaped cells known as guard cells which enclose stomatal pore. In grasses, the guard cells are dumb-bell shaped. The guard cells possess chloroplasts and regulate the opening and closing of stomata. Sometimes, a few epidermal cells, in the vicinity of the guard cells become specialised in their shape and size and are known as subsidiary cells. The stomatal aperture, guard cells and the surrounding subsidiary cells are together called stomatal apparatus. 'C' is the chloroplast. Since the functions of the mitochondria, chloroplast and peroxisomes are not coordinated with the above components, these are not considered as part of the endomembrane system.



Diagrammatic representation: (a) stomata with bean-shaped guard cells

(b) stomata with dumb-bell shaped guard cell

(NCERT Ed. 2025-26, 11th, Page No. 72)

[Contd...

100. (4)
In most higher plants, the growing apical bud inhibits the growth of the lateral (axillary) buds, a phenomenon called apical dominance. Removal of shoot tips (decapitation) usually results in the growth of lateral buds.
(NCERT Ed. 2025-26, 11th, Page No. 175)

101. (3)
Microvilli are minute finger-like projections of the plasma membrane found in animal cells (specifically epithelial cells like those in the intestine) to increase surface area for absorption. Plant cells possess a rigid cell wall outside their plasma membrane, which prevents the formation of microvilli.
(NCERT Ed. 2025-26, 11th, Page No. 100)

102. (1)
Mayr was awarded the three prizes widely regarded as the triple crown of biology: the Balzan Prize in 1983, the International Prize for Biology in 1994, and the Crafoord Prize in 1999.
(NCERT Ed. 2025-26, 11th, Page No. 2)

103. (1)
The LHC are made up of hundreds of pigment molecules bound to proteins. Each photosystem has all the pigments (except one molecule of chlorophyll *a*) forming a light harvesting system also called antennae. These pigments help to make photosynthesis more efficient by absorbing different wavelengths of light. The single chlorophyll *a* molecule forms the reaction centre.
(NCERT Ed. 2025-26, 11th, Page No. 138)

104. (2)
Glucose is partially oxidized in glycolysis. NADH + H^+ is formed from NAD^+ when 3-phosphoglyceraldehyde (PGAL) is converted to 1, 3-bisphosphoglycerate (BPGA).
(NCERT Ed. 2025-26, 11th, Page No. 156)

105. (3)
If the margins of sepals or petals overlap one another but not in any particular direction as in *Cassia* and gulmohur, the aestivation is called imbricate.
(NCERT Ed. 2025-26, 11th, Page No. 64)

106. (2)
The first word in a biological name represents the genus while the second component denotes the specific epithet.
(NCERT Ed. 2025-26, 11th, Page No. 4)

107. (1)
In diatoms the cell walls form two thin overlapping shells, which fit together as in a soap box. The cell wall (of Dinoflagellates) has stiff cellulose plates on the outer surface.
(NCERT Ed. 2025-26, 11th, Page No. 14)

108. (1)
Xylem parenchyma cells are living and thin-walled. In roots, the protoxylem lies towards periphery and metaxylem lies towards the centre.
(NCERT Ed. 2025-26, 11th, Page No. 87)

109. (4)
The parent cell has 16 chromosomes, and the daughter cells have 8 chromosomes. Since the chromosome number is reduced by half, the cell has undergone meiosis. The final daughter cells resulting from meiosis (haploid gametes) will have half the chromosome number and half the mass of DNA compared to the original diploid parent cell in the G_1 phase.
(NCERT Ed. 2025-26, 11th, Page No. 121)

110. (1)
Because of the specific restricted requirement of gametophytes (require cool, damp, shady places to grow) and the need for water for fertilisation, the spread of living pteridophytes is limited and restricted to narrow geographical regions
(NCERT Ed. 2025-26, 11th, Page No. 32)

111. (2)
Ethylenes enhances the respiration rate during ripening of the fruits. This rise in rate of respiration is called respiratory climactic. Gibberellins promote bolting. Abscisic acid is also called the stress hormone. Auxins cause apical dominance.
(NCERT Ed. 2025-26, 11th, Page No. 177)

112. (3)
During non-cyclic photophosphorylation NADPH + H^+ is synthesised. During this process, protons accumulate inside the thylakoid lumen due to the splitting of water and proton pumping. An increase in H^+ concentration causes the pH to decrease (become more acidic), not increase. Non-cyclic photophosphorylation (the Z-scheme) involves the coordination of both Photosystem II (PS II) and Photosystem I (PS I) connected by an electron transport chain. he electrons lost from PS I are compensated by electrons coming from PS II (via the electron transport chain). The electrons lost from PS II are the ones compensated by the splitting of water.
(NCERT Ed. 2025-26, 11th, Page No. 140)

[Contd...

113. (4)

The ATP synthase consists of two major components, F_1 and F_0. The F_1 headpiece is a peripheral membrane protein complex and contains the site for synthesis of ATP from ADP and inorganic phosphate. F_0 is an integral membrane protein complex that forms the channel through which protons cross the inner membrane. The passage of protons through the channel is coupled to the catalytic site of the F_1 component for the production of ATP.

(NCERT Ed. 2025-26, 11th, Page No. 161)

114. (1)

The genetic material is basically naked, not enveloped by a nuclear membrane. In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA called plasmids. A specialised differentiated form of cell membrane called mesosome is the characteristic of prokaryotes. No organelles, like the ones in eukaryotes, are found in prokaryotic cells except for ribosomes.

(NCERT Ed. 2025-26, 11th, Page No. 90)

115. (1)

A flower is a modified shoot wherein the shoot apical meristem changes to floral meristem. When a shoot tip transforms into a flower, it is always solitary.

(NCERT Ed. 2025-26, 11th, Page No. 61)

116. (3)

With the exception of yeasts which are unicellular, fungi are filamentous. A zygospore is formed by fusion of two gametes in members of phycomycetes E.g.; *Rhizopus* (the bread mould).

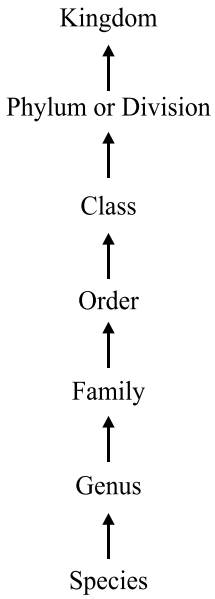
(NCERT Ed. 2025-26, 11th, Page No. 17)

117. (2)

Parenchyma cells are living and their walls are thin and made up of cellulose. Collenchyma consists of cells which are much thickened at the corners due to a deposition of cellulose, hemicellulose and pectin. Sclerenchyma cells are usually dead and without protoplasts. Xylem fibres have highly thickened walls and obliterated central lumens. These may either be septate or aseptate.

(NCERT Ed. 2025-26, 11th, Page No. 86)

118. (4)



Taxonomic categories showing hierarchial arrangement in ascending order

(NCERT Ed. 2025-26, 11th, Page No. 8)

119. (1)

Mitosis usually results in the production of diploid daughter cells with identical genetic complement. Cell growth results in disturbing the ratio between the nucleus and the cytoplasm. It therefore becomes essential for the cell to divide to restore the nucleo-cytoplasmic ratio.

(NCERT Ed. 2025-26, 11th, Page No. 125)

120. (3)

Unlike bryophytes and pteridophytes, in gymnosperms, the male and the female gametophytes do not have an independent free-living existence.

(NCERT Ed. 2025-26, 11th, Page No. 33)

121. (4)

Ethylene is used to initiate flowering and for synchronising fruit-set in pineapples.

(NCERT Ed. 2025-26, 11th, Page No. 177)

122. (2)

Carbon dioxide is the major limiting factor for photosynthesis.

(NCERT Ed. 2025-26, 11th, Page No. 150)

123. (3)

In fermentation, say by yeast, the incomplete oxidation of glucose is achieved under anaerobic conditions by sets of reactions where pyruvic acid is converted to CO_2 and ethanol. In animal cells also, like muscles during exercise, when oxygen is inadequate for cellular respiration pyruvic acid is reduced to lactic acid by lactate dehydrogenase. Phosphoglycolate and phosphoglycerate are products of photorespiration in C_3 plants.

(NCERT Ed. 2025-26, 11th, Page No. 157)

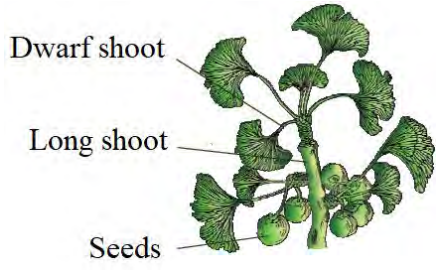
124. (1)
The major lipids of a plasma membrane are phospholipids that are arranged in a bilayer. The ratio of protein and lipid varies considerably in different cell types. In human beings, the membrane of the erythrocyte has approximately 52 per cent protein and 40 per cent lipids.
(NCERT Ed. 2025-26, 11th, Page No. 93)

125. (3)
Bacteria reproduce mainly by fission.
(NCERT Ed. 2025-26, 11th, Page No. 14)

126. (2)
In a dicot stem, cortical layers below hypodermis consist of rounded thin walled parenchymatous cells with conspicuous intercellular spaces. The innermost layer of the cortex is called the endodermis. The cells of the endodermis are rich in starch grains and the layer is also referred to as the starch sheath. In a dicot root, it comprises a single layer of barrel-shaped cells without any intercellular spaces. The tangential as well as radial walls of the endodermal cells have a deposition of water-impermeable, waxy material suberin in the form of casparian strips. Next to endodermis lies a few layers of thick-walled parenchymatous cells referred to as pericycle. Initiation of lateral roots and vascular cambium during the secondary growth takes place in these cells.
(NCERT Ed. 2025-26, 11th, Page No. 75)

127. (4)
The placentation in members of family Compositae is basal. The fruit of Compositae is a Cypsela. Compositae is a Dicotyledonous family. Therefore, the embryo has two cotyledons. The flowers (florets) in the head inflorescence (capitulum) are surrounded by a whorl of bracts.
(NCERT Ed. 2025-26, 11th, Page No. 65)

128. (1)
The given plant is *Ginkgo.* It is a gymnosperm. All gymnosperm are heterosporous. Its pollen grains are carried in air currents. The seeds are naked. The female gametophyte is multicellular.



Ginkgo
(NCERT Ed. 2025-26, 11th, Page No. 33)

129. (3)
The cells that do not divide further exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle.
(NCERT Ed. 2025-26, 11th, Page No. 122)

130. (4)
In fermentation there is a net gain of only two molecules of ATP for each molecule of glucose degraded to pyruvic acid.
(NCERT Ed. 2025-26, 11th, Page No. 162)

131. (2)
GA_3 is used to speed up the malting process in brewing industry.
(NCERT Ed. 2025-26, 11th, Page No. 176)

132. (4)
The number of NADPH required for completion of one Calvin cycle is 2.
(NCERT Ed. 2025-26, 11th, Page No. 144)

133. (1)
Slime moulds: Under suitable conditions, they form an aggregation called plasmodium which may grow and spread over several feet.
(NCERT Ed. 2025-26, 11th, Page No. 15)

134. (2)
The axoneme usually has nine doublets of radially arranged peripheral microtubules, and a pair of centrally located microtubules. Such an arrangement of axonemal microtubules is referred to as the 9+2 array.
(NCERT Ed. 2025-26, 11th, Page No. 99)

135. (3)
Axillary buds of stems may also get modified into woody, straight and pointed thorns. Thorns are found in many plants such as *Citrus*, *Bougainvillea*. Some plants of arid regions modify their stems into flattened (*Opuntia*), or fleshy cylindrical (*Euphorbia*) structures. A lateral branch with short internodes and each node bearing a rosette of leaves and a tuft of roots is found in aquatic plants like *Pistia* and *Eichhornia*. Stem tendrils which develop from axillary buds, are slender and spirally coiled and help plants to climb such as in gourds (cucumber, pumpkins, watermelon) and grapevines.
(NCERT Ed. 2025-26, 11th, Page No. 59)

136. (3)

Vertebrates are a subgroup of chordates, but all chordates do not possess vertebral columns.
the notochord is not replaced by vertebral column in all chordates; only vertebrates develop vertebrae, while in other chordates the notochord may persist.

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137. (3)

Secondary metabolites like rubber and many drugs have ecological roles.
Morphine and codeine are alkaloids.
Concanavalin A is a lectin, not a toxin.
Anthocyanins are plant pigments.
Cellulose and gums are polymeric carbohydrates.

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138. (3)

Tight junctions form a seal between adjacent cells and prevent leakage of substances across epithelial tissues, helping maintain tissue integrity and selective permeability.

(NCERT Ed. 2020-21, 11th, Page No. 102)

139. (3)

Basic amino acids have additional basic groups ($-NH_2$) in their side chain. Structure D shows a side chain ending with an amino group, which is characteristic of basic amino acids such as lysine.

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140. (1)

List-I (Wave)	List-II (Electrical Activity)
P-wave	Depolarisation of atria
QRS complex	Depolarisation of ventricles
T-wave	Repolarisation of ventricles
End of T-wave	End of ventricular systole

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141. (1)

Birds like *Pavo* excrete nitrogenous waste mainly as uric acid, which is eliminated in semisolid paste or pellet form (uricotelic excretion), helping conserve water.

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142. (2)

During contraction, the sarcomere shortens, so the distance between two Z-lines decreases.
During contraction, the sarcomere shortens, so the distance between two Z-lines decreases.
The I band (region of only thin filaments) actually decreases in length.
The H zone disappears during maximal contraction as actin filaments overlap the myosin filaments.

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143. (2)

The pneumotaxic centre, which helps moderate the activity of the respiratory rhythm centre, is located in the upper part of the pons. It regulates breathing by limiting inspiration duration and thus controls respiratory rate. The other options are not associated with pneumotaxic regulation.

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144. (3)

List-I	List-II
Alpha-cells of Islets of Langerhans	Glucagon; Glycogenolysis
Beta-cells of Islets of Langerhans	Insulin; Glycogenesis
Zona glomerulosa	Aldosterone; Electrolyte balance
Zona fasciculata	Cortisol; Anti-inflammatory

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145. (3)

The resting axonal membrane has very low permeability to Na^+ ions and large negatively charged proteins inside the axoplasm, helping maintain resting membrane potential.
The active transport mechanism of the Sodium-potassium pump moves 3 Na^+ ions outwards and 2 K^+ ions inwards, not 2 Na^+ outwards and 3 K^+ inwards.

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146. (2)

Members of phylum Mollusca have a body divided into head, muscular foot, and visceral hump, which is a characteristic body plan of molluscs. Other options do not show this specific division.

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147. (3)

In a typical cell, nucleic acids (DNA and RNA) contribute only a small fraction of the total cellular mass, approximately 5–7%, whereas proteins form the major component of cellular dry mass.

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148. (4)

Adamsia (Sea anemone) is a Cnidarian and lacks radula (feeding organs found in Mollusca). *Obelia* belongs to the group that shows metagenesis/polymorphism.

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149. (2)

Cardiac output = Stroke volume × Heart rate
= 50 ml × 100 beats per minute
= 5000 ml (or 5 L) per minute.
This represents the volume of blood pumped per minute by the Heart.

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150. (1)

Competitive inhibitors are structurally similar to the substrate.

Because of structural similarity, competitive inhibitors bind to the active site of the Enzyme, blocking substrate binding and inhibiting enzyme activity.

(NCERT Ed. 2025-26, 11th, Page No. 117)

151. (2)

The larynx is a cartilaginous box which helps in sound production and is hence called the sound box.

(NCERT Ed. 2025-26, 11th, Page No. 184)

152. (2)

Most renal calculi (kidney stones) are composed of calcium oxalate crystals, sometimes mixed with phosphate. Other salts like sodium chloride, potassium carbonate, or magnesium sulphate are rarely involved in stone formation.

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153. (2)

The Cranium consists of 8 bones.

The human skull is dicondylic, having two occipital condyles articulating with the vertebral column.

Ribs 8–10 are considered false ribs.

The Hyoid bone is U-shaped and located at the base of the buccal cavity.

The middle ear contains three ossicles: Malleus, Incus, and Stapes.

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154. (2)

Oxytocin acts on the uterine muscles and causes stronger uterine contractions, which in turn stimulates further secretion of oxytocin (positive feedback).

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155. (4)

Nissl bodies are large granular structures in the cytoplasm of neurons and are mainly composed of free ribosomes and rough endoplasmic reticulum (RER). They are responsible for active protein synthesis in the Neuron.

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156. (1)

The primary structure of a protein is the linear sequence of amino acids in the Protein.

The first amino acid is the N-terminal, not C-terminal.

Proteins usually form right-handed α-helices, not only left-handed helices.

The tertiary structure is essential for the biological activity of most proteins.

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157. (2)

VC is the maximum volume of air a person can breathe in after a forced expiration (TV+IRV+ERV). TV is 500ml. RV is 1100-1200ml. IC is TV+IRV.

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158. (4)

In an open circulatory system, blood is not circulated through a completely closed network of blood vessels; instead, it flows into open spaces called hemocoel.

Both Annelida and Chordata possess a closed circulatory system, where blood flows inside blood vessels.

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159. (2)

The dialysing fluid used in hemodialysis is designed to have a composition similar to plasma, but it lacks large proteins, so that waste products diffuse out from the blood. This process is used to filter blood in patients with kidney failure involving the Kidney.

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160. (3)

The mechanism of muscle contraction is explained by the Sliding filament theory.

Contraction occurs by sliding of thin (actin) filaments over thick (myosin) filaments, not thick over thin filaments.

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161. (2)

Ribose is a monosaccharide pentose sugar present in RNA. It differs from deoxyribose, which is found in DNA. Glucose and fructose are hexose sugars and are not components of RNA.

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162. (1)

The Juxtaglomerular apparatus is formed by modified cells of the distal convoluted tubule and the afferent arteriole.

A decrease in Glomerular filtration rate stimulates the juxtaglomerular (JG) cells to release Renin, which helps restore blood pressure and filtration rate.

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163. (2)

The Corpus callosum is a large bundle of nerve fibres that connects the left and right cerebral hemispheres and is located in the forebrain region. It facilitates communication between the two hemispheres.

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164. (2)

List-I (Hormone)	List-II (Nature)
Insulin	Peptide/Protein
Epinephrine	Emergency hormone and amino-acid derivative
Testosterone	Steroid
Thyroxine	Iodothyronine

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165. (2)

The Hilum of kidney is a notch on the medial (inner concave) border of the kidney through which structures like the renal artery, renal vein, and ureter pass.

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166. (3)

White muscle fibre (also called fast-twitch fibres) have low myoglobin content, fewer mitochondria, and mainly depend on anaerobic metabolism for quick, powerful contractions. They contain a well-developed sarcoplasmic reticulum, which helps in rapid release and uptake of calcium ions during contraction.

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167. (3)

RBCs contain a very high concentration of the enzyme, carbonic anhydrase and minute quantities of the same is present in the plasma too.

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168. (2)

Among the white blood cells, basophils are the least abundant, constituting about 0.5–1% of total Leukocyte count in normal blood. They are involved in inflammatory and allergic responses. Other WBCs like neutrophils and lymphocytes are more abundant.

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169. (3)

Ammonotelism is the excretion of nitrogen mainly as ammonia, which is common in aquatic animals because ammonia is highly toxic but easily diluted in water.

Bony fishes, aquatic insects (larval forms), and aquatic amphibians are generally ammonotelic.

Reptiles are uricotelic, excreting nitrogen mainly as uric acid to conserve water

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170. (4)

Prawns use Antennal glands (Green glands) for excretion. Statocysts are balance organs.

(NCERT Ed. 2020-21, 11th, Page No. 103)

171. (2)

List-I (Phylum)	List-II (Feature)
Ctenophora	Comb plates
Mollusca	Radula
Porifera	Choanocytes
Echinodermata	Water vascular system

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172. (1)

Almost all mammals have seven cervical vertebrae, with rare exceptions (e.g., sloth, manatee).

Proteins are heteropolymers made from 20 different types of amino acids.

Heart failure (reduced pumping efficiency) is different from cardiac arrest (sudden stoppage of heart).

Inulin is a polymer of fructose.

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173. (3)

DCT is the site for conditional reabsorption of Na^+ and water, often mediated by hormones like Aldosterone and ADH.

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174. (2)

Human urine is slightly acidic with an average pH of 6.0.

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175. (1)

List-I (Protein)	List-II (Function)
Collagen	Intercellular ground substance
Trypsin	Enzyme
Insulin	Hormone
Receptor	Sensory reception

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176. (2)

Taenia (Platyhelminthes) is an acoelomate. *Ascaris* is a pseudocoelomate. *Nereis* and *Pila* are coelomates.

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177. (1)

- In the PNS, myelin sheath is produced by Schwann cells.
- In the CNS, it is produced by Oligodendrocytes.
- Astrocytes provide support and nutrition, while osteoclasts are bone-resorbing cells and not involved in myelination.

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178. (3)

In cockroach (*Periplaneta americana*), only a small part of the nervous system is present in the head (supra-oesophageal ganglion), while the major part runs along the ventral side as a double ventral nerve cord with segmental ganglia.

In cockroach, 6–8 hepatic (gastric) caeca arise at the junction of foregut and midgut, not at the junction of midgut and hindgut.

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179. (1)

The human heart is myogenic, meaning the heartbeat originates from the heart muscle itself.

The normal rhythmic activity is generated intrinsically by specialised cardiac muscle cells (nodal tissue), particularly the Sinoatrial node, which acts as the pacemaker.

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180. (3)

Frogs respire in water through their skin (cutaneous respiration) by diffusion of dissolved oxygen. On land, they use skin, buccal cavity, and lungs (pulmonary respiration)

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